「Def. Given a space X, a covering space of X consists of a space $\tilde{X}$ and a map $p: \tilde{X} \to X$ s.t.
For each $x \in X$, there is an open neighborhood U of x in X satisfying

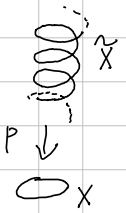
$$p^{-1}[U] = \coprod_{i \in \Lambda} \mathcal{O}_i \quad \text{and} \quad p|_{\mathcal{O}_i}: \mathcal{O}_i \xrightarrow{\cong} U_i.$$

(Such U is called "evenly covered"

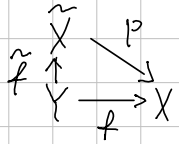
「Example. Given a space S^1, let $p: \mathbb{R} \to S^1$; $s \mapsto (\cos 2\pi s, \sin 2\pi s)$.
Let $(1,0) \in S^1$ be given. Then, for a neighborhood $B_1((1,0)) \cap S^1$,

$$p^{-1}[B_1((1,0)) \cap S^1] = \coprod_{n \in \mathbb{Z}} \left(n + \left(-\tfrac{1}{6}, \tfrac{1}{6}\right)\right)$$

and for each $n \in \mathbb{Z}$,
$\left(n + \left(-\frac{1}{6}, \frac{1}{6}\right)\right) \cong B_1((1,0)) \cap S^1$ via p.



「Def. Given a covering $p: \tilde{X} \to X$ and a map $f: Y \to X$,
a continuous map $\tilde{f}: Y \to \tilde{X}$ such that $p \circ \tilde{f} = f$ is called a lift of f to $\tilde{X}$,
with respect to p.



Thm. Given a map $F: Y \times I \to X$ and a map $\tilde{F}: Y \times \{0\} \to \tilde{X}$ lifting $F|_{Y \times \{0\}}$, there is a unique map $\tilde{F}: Y \times I \to \tilde{X}$ lifting F extended $\tilde{F}: Y \times \{0\} \to \tilde{X}$.

Proof.

Def. Let $p: E \to B$ be a covering map.
Given a continuous map $f: X \to B$, a lifting $\tilde{f}: X \to E$ of f is a map satisfying $p \circ \tilde{f} = f$.

Thm. Let $p: E \to B$ be a covering and $p(e_0) = b_0$.
For any path $f: [0,1] \to B$ with $f(0) = b_0$,
there exists a unique path $\tilde{f}: [0,1] \to E$ satisfying $p \circ \tilde{f} = f$
Proof, 보아들인다.
For any associated map by Homotopy of paths $F: [0,1] \times [0,1] \to B$ with $F(0,0) = b_0$,
there exists a unique associated map by Homotopy of paths $\tilde{F}: [0,1] \times [0,1] \to E$
satisfying $p \circ \tilde{F} = F$ and $\tilde{F}(0,0) = e_0$.